

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

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Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity descending a squared-drift objective preserves coherence while moving away from the environmental structure it should learn, below a shuffled-label comparison: dissipation minimization does not imply representational learning. A Lean 4 formalization composes finite capacity, Landauer’s

bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes; a common toy model satisfies all eight. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

Such an account has three requirements, and they order what follows. Coupling must be strong enough to sustain coherent collective activity; the local descriptions that activity carries must be compatible enough to glue into a single one; and that glued state must support an accurate model of itself. The first requirement is physical and testable with current methods, and most of this article is about testing it. The second and third are where

17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset expo-
34 nent measured from a finite ramp is a function of ramp speed across the
35 sweep of Section 4, so an exponent fitted to a recovery trace measures the
36 ramp at least as much as the mechanism—and that holds for any analysis
37 reading a critical exponent off a driven transition, not only for ours. The
38 third is a negative result about plasticity: a coupling update descending
39 a squared-drift objective preserves coherence while moving *away* from the

40 environmental structure it is expected to learn, below a shuffled-label com-
41 parison. Dissipation minimization does not imply representational learning.
42 The fourth is a machine-checked composition (Section 5) in which finite in-
43 formation capacity, Landauer’s bound, a predictive dissipation bound, the
44 noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point theorem
45 yield one conditional theorem, with the eight hypotheses that composition
46 consumes isolated and named; a common toy model satisfies all eight, so
47 their conjunction is not empty.

48 A prose table of assumptions would list the same eight items, so the fourth
49 contribution needs a word about method. What the Lean development adds
50 is that it makes the list non-negotiable. The characteristic failure in an
51 argument of this shape is not a false step but a silent change of subject:
52 a theorem about a stationary density read as a claim about a trajectory,
53 an inequality about a register read as a claim about learning, agreement of
54 phases read as compatibility of representations. Type-checking forbids each
55 of those substitutions, and an assumption needed to make a step go through
56 appears as a hypothesis in the term rather than as a word in a sentence. The
57 last of them is not merely forbidden but refuted by a constructed object: the
58 development carries a cover at a single common phase whose local densities
59 disagree where its patches meet, satisfying every other requirement of the
60 gluing theorem and failing compatibility alone (Section 5.2). That object
61 is what a prose table of assumptions cannot carry. Formalization does not
62 make the assumptions true; it makes them impossible to lose.

63 The physical model and the formal composition belong in one article be-
64 cause they are two halves of one commitment. The coupling-gated prediction
65 is worth testing under this framework, rather than as one more synchroniza-
66 tion model, only because the coherent state it produces is claimed to do
67 further work: to support a globally consistent representation and a reflexive
68 fixed point that we identify with phenomenal unity and a minimal self.

69 The distinction that does that work is between phases agreeing and de-
70 scriptions fitting together, and it is concrete. Take two cortical regions whose
71 receptive territories overlap, each carrying a probability distribution over
72 what occupies the shared part. Their oscillators can be locked to the same
73 phase at every instant while one region's distribution on the overlap places an
74 object left of fixation and the other places it right. The phases agree; the de-
75 scriptions do not, and no single distribution over the shared territory restricts
76 to both. A theory that identifies unity with synchrony cannot mark that dif-
77 ference, and so cannot separate a brain whose regions oscillate together from
78 one whose regions are about the same world. The sheaf condition marks
79 exactly it: local states glue to one global section when their restrictions to
80 shared regions coincide, and otherwise not at all. Coherence enters as the
81 physical condition under which compatibility can be maintained, not as a
82 substitute for it, and the self-prediction map then asks the further ques-
83 tion of whether the glued state contains an accurate model of itself. That
84 identification is a commitment, not a theorem, and Section 6 states it, ar-
85 gues for it, and says what would count against it. Its separation from the

physical tests is what makes both assessable: the cortical coupling hypothesis and the recovery prediction can fail while the mathematical composition stands, and the composition can stand while the identification is wrong. The established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988; Banach, 1922) therefore appear here for their biological composition, not for any new fundamental physical law.

2 A coupling-gated model of cortical coherence

2.1 Dynamics and stationary solutions

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles treated as dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are separate parameters; a frequency-spread width cannot be substituted for D when applying the identical-frequency stationary theory.

For identical frequencies in their rotating frame, the assumed stationary

102 density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

103 The I_n are modified Bessel functions, and ψ is the collective phase. Lean
 104 verifies normalization, the density's circular order parameter, and the self-
 105 consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent
 106 order parameter is zero. At $K > K_c$, exactly one positive solution coex-
 107 ists with zero; the positive branch is continuous at threshold and strictly
 108 increasing with K .

109 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

110 This is a result about the stationary solution branch. Of the coherent branch
 111 we prove existence, uniqueness, continuous emergence at threshold and strict
 112 growth in K , but not that it is dynamically selected. The dynamical mean-
 113 field limit—propagation of chaos—is the missing link between the finite sys-
 114 tem and the stationary description (Sznitman, 1991), and no theorem in this
 115 development closes it. Exchange symmetry alone does not justify factoriza-
 116 tion of the joint law, particularly for an ensemble with an unpinned collective
 117 phase.

118 A spatial version replaces the finite sum by an integral against a kernel
 119 $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-

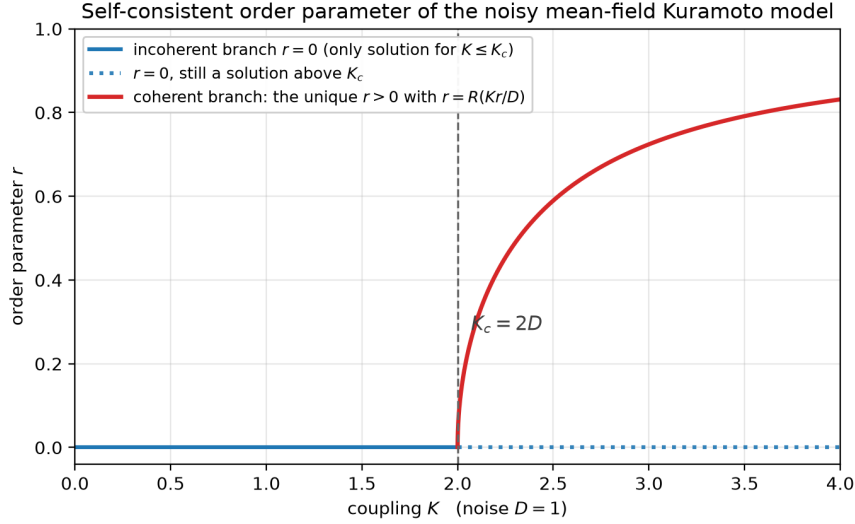


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

field strength is a kernel average, not a quantity that grows with the arbitrary
measure of the substrate. Reduction of a heterogeneous spatial kernel to a
scalar K is an additional modelling step, and the threshold in Eq. (2) does
not automatically apply to every spatial or signed network.

2.2 Physical identification and calibration

Regular mesh refinement gives convergence of a discrete coupling-energy
quadrature to its continuum integral: scalar energy convergence, not conver-
gence of trajectories and not a construction of a physical kernel. The model

128 posits a continuum kernel and separately identifies its cortical realization
 129 with endogenous electromagnetic coupling. The formal predicate requires
 130 a jointly continuous kernel, a normalized substrate, a mean-field strength
 131 equal to its spatial average, and no single site providing more than half of
 132 that mean-field coupling. Toy witnesses satisfy these conditions; none is a
 133 model calibrated to cortex.

134 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
 135 2010; Anastassiou & Koch, 2015), and field-based models have been proposed
 136 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
 137 Fadden, 2020). These observations motivate an effective interaction without
 138 establishing its magnitude relative to phase diffusion, its spatial averaging
 139 rule, or its causal role in experience; changes in phase organization under
 140 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
 141 coupling threshold.

142 The physical-unit conversion is the sharpest open limitation of the frame-
 143 work. An estimate proportional to $N\Delta tf$, using population size, a spike-time
 144 shift and oscillation frequency, is dimensionless. It cannot be compared di-
 145 rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

146 requires a rate calibration γ , where s is the time shift per unit field and E
 147 the field amplitude. Supplemental Material sets out why γ is not recoverable

148 from the other factors: it states how a per-cycle phase advance accumulates
 149 into a drift rate, and the choice that would let it cancel removes the very
 150 ratio the threshold test is about. The saved field-strength calculations are
 151 conditional on this calibration and on the chosen spatial aggregation; they do
 152 not independently establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction
 153 kernel and phase diffusion from the same physical system is necessary before
 154 the threshold becomes a quantitative biological claim; Section 3.2 makes that
 155 measurement the first item of the proposed experiment.

156 **3 Recovery after awakening: a conditional** 157 **prediction**

158 Sleep-associated changes in extracellular space and clearance motivate a pos-
 159 sible slow control of coupling (Xie et al., 2013). We hypothesize that recovery
 160 after awakening includes a change in effective coupling $K(t)$, mediated by
 161 extracellular geometry, on a slower timescale than phase dynamics. Neither
 162 the monotonic relation between geometry and coupling nor its role in hu-
 163 man sleep inertia follows from the cited physiological observation; the model
 164 requires both to be measured.

165 If $K(t)$ rises through K_c and the phase distribution tracks the coherent

166 stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

167 The ideal stationary description has a subthreshold interval followed by con-
 168 tinuous onset. Saturation requires a specified subsequent coupling trajectory;
 169 monotonic coupling alone does not determine a logistic or any unique sig-
 170 moid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently
 171 known. When coupling is inferred from the same coherence trace, this is not
 172 a parameter-free prediction of latency.

173 Critical slowing down limits the adiabatic prediction of Eq. (5) near
 174 threshold, and a finite population carries nonzero fluctuating order below
 175 it. An observable onset therefore requires a predeclared threshold above the
 176 finite-size floor, and a finite-rate trajectory need not show an infinite-slope
 177 cusp. Section 4.1 quantifies the limitation.

178 **3.1 Distributional and temporal tests**

179 The stationary model makes a second prediction: phase concentration and
 180 coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its appli-
 181 cability depends on the stationary-density assumption and the observation
 182 model, and the relation is common to the whole von Mises family, so agree-
 183 ment with it identifies neither ephaptic coupling nor this framework against
 184 alternative synchronization models. Within the same ansatz $a/r = K/D$ for

185 $r > 0$, so the limiting threshold ratio follows from the density relation instead
186 of testing it, and ratios near the finite-size floor are especially unstable.

187 The estimators must be independent, which is the constraint most easily
188 violated in practice. Estimating a by inverting the Bessel ratio from r would
189 enforce the relation being tested, and linear statistics on wrapped phases
190 estimate neither quantity.

191 **3.2 A protocol for the awakening experiment**

192 The requirements above define a design rather than a single measurement,
193 and the design is where the content of the hypothesis lives. An adequate test
194 needs all six of the following.

- 195 **1. Coupling and diffusion in the same units.** Estimate K and D
196 from the same cortical regime, in inverse-time units, separating phase
197 diffusion from quenched frequency spread and from observation noise.
198 Without this the threshold $K > 2D$ is not an evaluable statement
199 about cortex.
- 200 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
201 an astrocytic signal is an indicator of extracellular geometry, not a mea-
202 surement of K . Its mapping to effective coupling must be established
203 separately, with its own uncertainty.
- 204 **3. Phase estimation at adequate spatial resolution.** At least 100
205 spatial samples per estimate. Estimate r by the circular resultant

length and a by an unweighted regression of log-density on the cosine of phase relative to the circular mean. With fewer sites, pooling over time changes the estimand to a spatiotemporal distribution and must be stated as such.

4. **An independent recovery endpoint.** Behavioral recovery measured separately from the phase trace, so that latency is not defined by the signal whose latency is at issue.

5. **A predeclared onset criterion.** A coherence threshold fixed above the finite-size floor before unblinding, chosen in the knowledge that a finite ramp smears the cusp.

6. **Model comparison under matched observation.** Coupling-gated recovery compared against exponential, flexible sigmoidal and other slow-recovery models under identical temporal filtering and observation noise. Latency alone cannot distinguish the mechanism, because many processes create delays.

The design fails informatively in three ways: failure of the stationary phase-density relation over a sufficiently broad concentration range; absence of the proposed coupling–geometry association; or a recovery trace better explained by an alternative after finite-rate and observation effects are accounted for. Two apparent failures are not failures. A linear apparent onset is evidence against the adiabatic approximation only where that approximation is independently justified, and it does not falsify the field hypothesis

228 under arbitrary ramp speeds. The absence of a fold in the stationary equa-
 229 tion excludes stationary coexistence of two positive branches in this model,
 230 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

231 4 Simulation results

232 The simulations characterize the finite-size, finite-rate regime that the sta-
 233 tionary theory leaves open, and they bear on whether the prediction of Sec-
 234 tion 3 is measurable at all. Two of them, Sections 4.1 and 4.3, illustrate
 235 methodological issues that also arise in analyses of driven transitions and
 236 dissipation-driven plasticity. Each is a finite run at its sampled parameters
 237 and not a general statement, and none discharges a hypothesis of the com-
 238 position in Section 5. Each run uses explicit seeds and compact saved sum-
 239 maries, and the numerical values below are generated from those artifacts.
 240 The supplement gives parameters, estimators, failure criteria and additional
 241 panels. Identical-frequency runs use phase diffusion rather than a quenched
 242 frequency-spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

243 4.1 Ramp speed sets the apparent onset exponent

244 The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at
 245 four coupling speeds. Escape is the first of three consecutive saved samples at
 246 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
 247 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,

248 with a fitted delay exponent 0.443. The apparent onset exponent changes
249 from 0.964 at the fastest speed to 0.417 at the slowest.

250 The exponent recovered by fitting a driven transition is a joint function
251 of the mechanism and the drive rate. An onset exponent read off a recovery
252 trace therefore says little about the underlying transition unless the ramp rate
253 is independently known and the finite-size floor is declared, and that holds
254 for any such analysis, not only for one conducted under this framework. The
255 prediction of Section 3 is the first thing it constrains: the stationary square-
256 root law cannot be imposed on an arbitrary recovery trace, and its absence
257 is not by itself evidence against a coupling-gated mechanism. The specified
258 deviation-based critical ramp speed is not bracketed by the sweep.

259 The dynamical-selection control shows escape toward a coherent branch
260 above threshold, but retains quantitative finite-size discrepancies. At $K =$
261 2.8, steady mean order is 0.67735 against the static value 0.68270. The
262 fitted growth-rate slope is 0.43240, compared with the theoretical slope $1/2$.
263 The propagation-of-chaos diagnostics show decreasing subcritical dependence
264 with increasing population size. In the supercritical unpinned ensemble,
265 however, pair correlation remains substantial because realizations share a
266 random collective orientation. This negative unconditional result does not
267 settle a conditional mean-field limit after removal of that orientation.

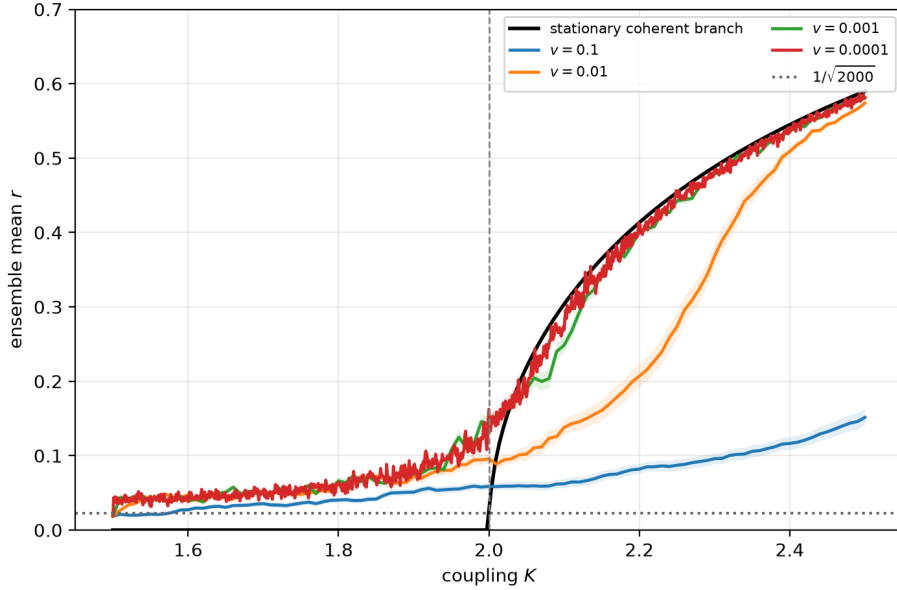


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

268 4.2 Spatial coupling, frustration and calibration

269 A periodic two-dimensional sheet with row-normalized exponential coupling
 270 remains coherent in the sampled empirical decay-length band under the de-
 271 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
 272 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
 273 eter regime; fixing total coupling while changing range tests fragmentation,
 274 not whether cortex supplies that total coupling.

275 A row-balanced Dale-law network shows that row balance alone does not

276 guarantee a floor-level incoherent baseline. In a separately specified weak-
 277 synapse regime, a uniform modulatory term produces an operational coher-
 278 ence crossing at K_{eff}/D between 1.8994 and 1.9387. That ratio is dimension-
 279 less; carrying it over to cortex needs the rate calibration γ of Section 2.2,
 280 which neither this sweep nor the spatial control supplies.

281 4.3 Squared-drift plasticity does not learn structure

282 A coupled system that descends a dissipation-like objective is widely expected
 283 to come, in doing so, to mirror the structure of its input; the predictive
 284 dissipation bound of Section 5 invites that reading. This control tests the
 285 expectation directly in the present model, and the expectation fails.

286 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

287 It is a quadratic functional of an affine image of the kernel. The formalization
 288 constructs the bounded drift operator A and its adjoint on the relevant L^2
 289 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
 290 decreases this objective at fixed phases. The objective is not generally a
 291 thermodynamic entropy-production rate, which requires probability-current
 292 information. Whether biological couplings follow this flow is an additional
 293 modelling question.

294 The joint control evolves noisy phases with clustered frequencies and

nonzero grand-mean drive, while updating symmetric couplings every 50 fast steps. Couplings remain nonnegative, hollow and normalized to a fixed total. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so both endpoints contribute. Resource constraints and evolving phases remove the fixed-phase monotonicity guarantee.

Across three seeds, second-half order lies between 0.9811 and 0.9815, and matched frozen-coupling controls maintain similar coherence. The distance to the true cluster template increases in every adaptive run, with worse changes than the shuffled-label comparison (Fig. 3). The template is a covariance of shared latent cluster signals, not an input to the update or a measured covariance of the constant frequencies.

The objective descends, coherence is maintained, and the coupling matrix moves away from the generating structure—below the level of a random relabelling. Coherence is therefore not a proxy for representational alignment in this regime, and a plateau in the objective establishes neither stationarity of the adaptive process nor a thermodynamic entropy-production rate.

4.4 The estimation protocol demonstrated on public EEG

Item 3 of Section 3.2 specifies estimators whose behavior on real recordings is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa et al., 2025). The purpose is to show what the protocol produces and where

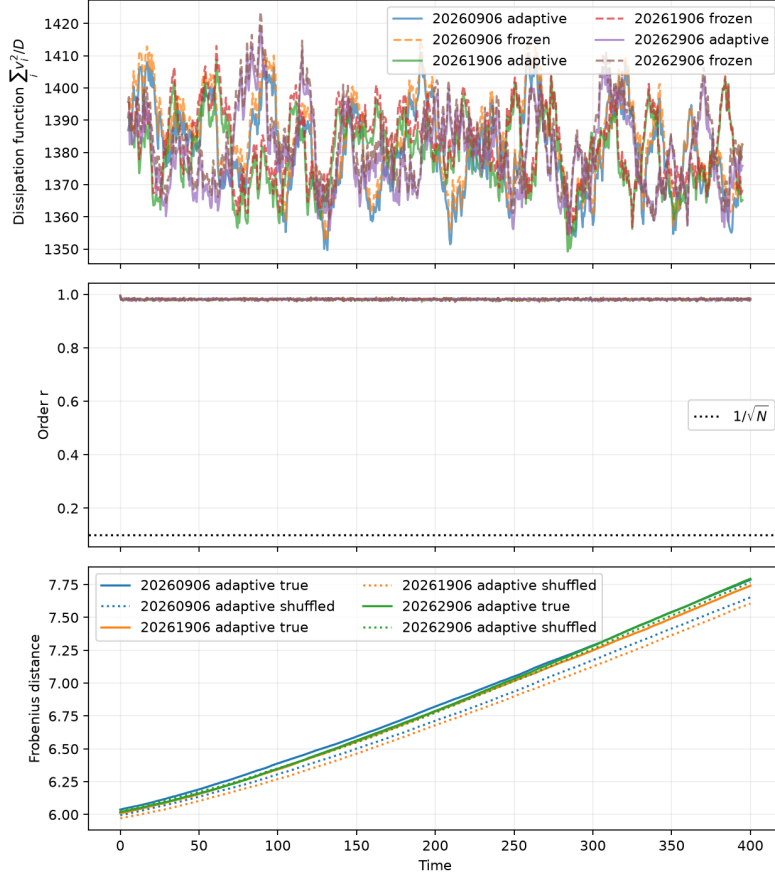


Figure 3: Joint fast/slow dynamics with matched frozen-coupling controls. Top: smoothed squared-drift dissipation, with frozen controls dashed. Middle: order and the finite-size reference. Bottom: distances to true cluster templates (solid) and node-label shuffles (dotted). Coherence persists, but structural distances increase. The result limits the proposed connection between dissipation minimization and environmental structure.

316 it is fragile, not to test the awakening hypothesis, for which these data are
317 the wrong design.

318 Two fragilities appear immediately. The implemented estimator pools 62
319 channels over a 100-ms window, so it estimates a spatiotemporal distribu-
320 tion rather than an instantaneous spatial one—a departure from the protocol
321 forced by channel count, and one that any scalp implementation will face.
322 Bipolar and common-reference montages give different residuals, making ref-
323 erencing a material analysis choice rather than a preprocessing detail.

324 The bipolar results are compatible with the Bessel relation over the ob-
325 served concentration range. That range is too narrow to distinguish the
326 curve from its linear approximation at the observed scatter, so the check is
327 uninformative about the relation itself. An informative test needs a broader
328 concentration range and explicit treatment of source mixing, finite samples
329 and temporal pooling; intracranial or high-density recordings across a gen-
330 uine state transition are the natural setting.

331 5 Thermodynamic constraints and the for- 332 mal composition

333 The synchronization model of Section 2 is one link of a longer conditional
334 construction, whose nine nodes are:

335 **n1** a finite phase space has bounded information capacity; **n2** a
336 broken symmetry forces a boundary, where the field leaves its vac-

uum manifold; **n3** the boundary erases, and erasure costs heat;
n4 that dissipation bounds nonpredictive memory; **n5** discrete
couplings coarse-grain to a continuum limit; **n6** a specified phys-
ical field realizes that limiting coupling; **n7** above threshold the
field carries a unique coherent order parameter; **n8** compatible
local states glue to a unique global section; **n9** that section is the
unique fixed point of a self-prediction map.

Node-level results are established or cited; what the composition consumes
are the eight implications between them. Figure 4 shows where they sit and
what kind of assumption each one is, and Table 1 states them, classified by
their role in the composition rather than by their plausibility. In particular,
finite capacity does not imply a symmetry-breaking boundary: n1 and n2 are
independent roots, even though the composition can accept an implication
between them as an argument.

5.1 Capacity, boundaries and predictive memory

For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
equality only for the uniform distribution. A stream with more possible
histories than $|X|$ cannot be encoded injectively in that register. Separately,
a continuous field on a connected substrate that reaches distinct components
of a disconnected vacuum manifold must leave that manifold somewhere,
the simplest topological-defect construction (Kibble, 1976). Relating this
locus to a particular finite information-processing register is the additional
hypothesis E12 carries.

A non-surjective update on a finite register is non-injective. Under the

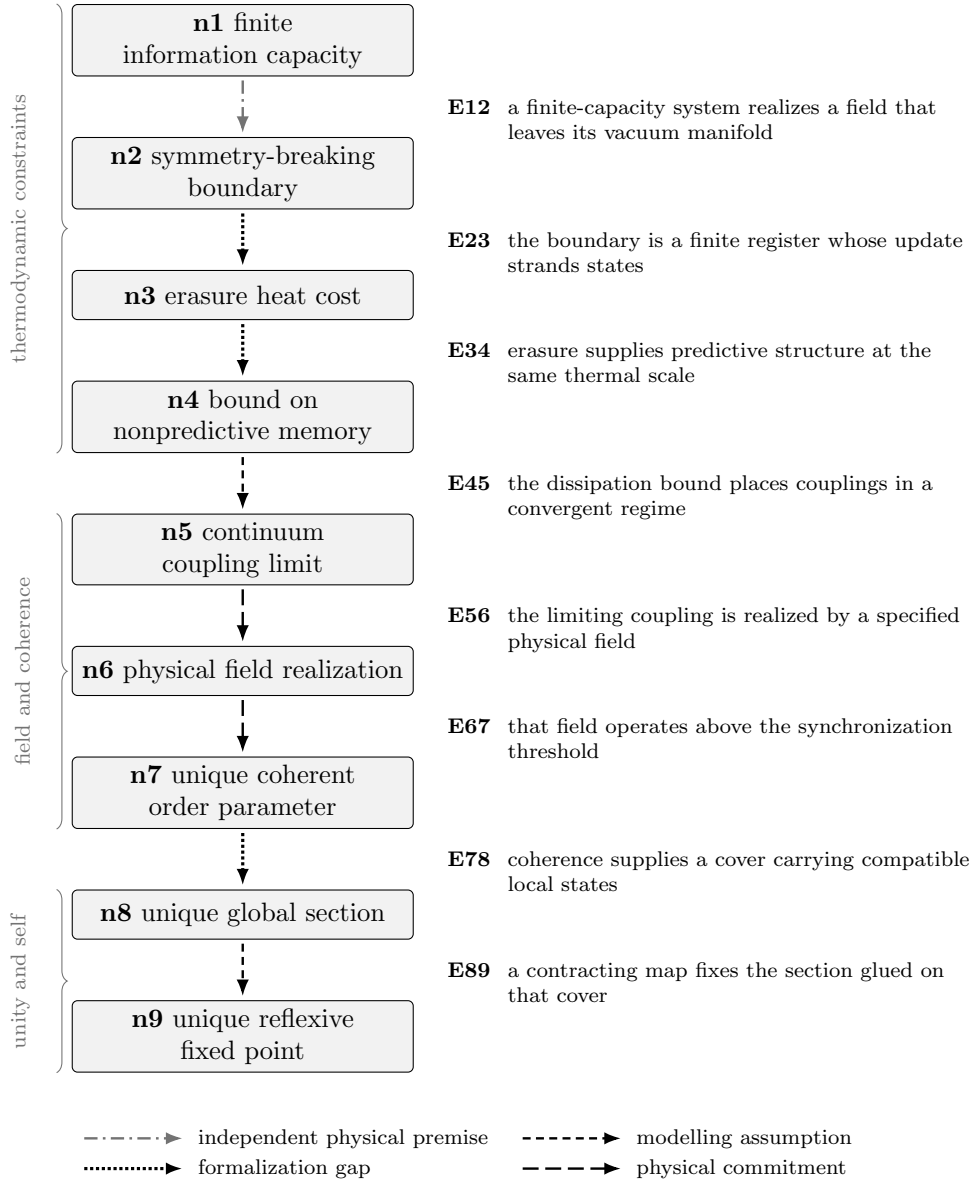


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

Edge	Status	Required connection
E12 capacity $\rightarrow$ boundary	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
E23 boundary $\rightarrow$ erasure	Formalization gap	The boundary is associated with a finite register whose update leaves states unreachable.
E34 erasure $\rightarrow$ prediction	Formalization gap	Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45 prediction $\rightarrow$ continuum	Modelling assumption	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56 continuum $\rightarrow$ field	Physical commitment	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
E67 field $\rightarrow$ coherence	Physical commitment	That field operates above the synchronization threshold.
E78 coherence $\rightarrow$ unity	Formalization gap	Coherence supplies a cover with coupling at least the named mean-field value, reached by relaxation and carrying compatible local states.
E89 unity $\rightarrow$ self	Modelling assumption	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

361 specified system–bath model and thermal assumptions, erasure carries a Lan-
 362 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
 363 update erases, and logical many-to-one processing must be distinguished from
 364 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 365 tion records the physical heat identity as an obligation of each model and
 366 supplies an explicit one-bit eraser satisfying it.

367 For a system state X_t driven by an environmental signal S_t , the predictive
 368 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

369 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
 370 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains
 371 nonpredictive memory at a given dissipation budget. It does not specify
 372 a learning rule, maximize prediction, or require the internal representation
 373 to reproduce environmental geometry—the point the plasticity control of
 374 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
 375 ditionally requires nonpredictive information to be bounded by the entropy
 376 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
 377 bound under that identification; it does not derive the identification.

378 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
 379 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
 380 tropy, not a general identification of relative entropy with physical entropy

381 production. These distinctions prevent an informational inequality from be-
382 ing read as a derivation of environmental learning, Bayesian inference, or
383 semantic content.

384 5.2 Compatible states and reflexive fixed points

385 A sheaf formalizes local descriptions and their restriction to smaller regions.
386 If local sections agree on every overlap, the sheaf condition gives a unique
387 global section. In the probability-sheaf construction, this is consistency of
388 local density information, and it is not implied by agreement of oscillator
389 phases: the development includes a counterexample with equal phases but
390 incompatible densities on an overlap. Section 6 rests on this distinction,
391 and the model excludes such configurations only through its compatibility
392 hypothesis.

393 A thermodynamic cover couples this gluing construction to a phase con-
394 figuration at a potential minimum. For the deterministic positive-coupling
395 model, convergence to that minimum is proved under an explicit initial basin
396 condition; other equilibria prevent an unconditional conclusion. The station-
397 ary noisy order parameter, the deterministic relaxation, and the cover must
398 therefore be connected by E78 rather than identified without a hypothesis.
399 Neither field continuity nor synchronization alone guarantees a globally com-
400 patible representation.

401 A reflexive extension equips global sections with a complete metric and a
402 predictive map formed by restricting a state to a local encoding region and

403 reading that encoding out. If the map contracts and fixes the section supplied
404 by the cover, Banach’s theorem gives uniqueness of that same fixed point.
405 The contraction rate is assumed, and the equality between the glued section
406 and the fixed point is part of E89. A coherent order parameter supplies the
407 supercritical inequality required by the stated contraction bound; it does not
408 construct the map or prove that cortical dynamics implements it.

409 The theorem `chain` concludes `UnifiedSelf` from these hypotheses. A
410 common toy witness satisfies all eight, so their conjunction is not empty.
411 The project declares no additional Lean axioms beyond the foundational
412 axioms used by Lean’s libraries; physical content resides in hypotheses and
413 model fields. This checks the conditional mathematical composition, not the
414 empirical truth of its premises. Detailed constructions, scope statements and
415 counterexamples are retained in the supplement and source files.

416 6 What the construction commits to about 417 consciousness

418 The composition terminates in a mathematical object, and we take a posi-
419 tion on what that object is. The commitment has two clauses. *Unity*: the
420 unique global section glued from compatible local states is the formal corre-
421 late of the unity of a conscious episode—the fact that a subject undergoes
422 one experience rather than a collection of co-occurring local ones. *Selfhood*:
423 the fixed point of the self-prediction map on that same section is the formal

424 correlate of a minimal self, a state that is what it takes itself to be. Neither
425 clause is a theorem, and no simulation here supports either. We argue for
426 them, then say what would count against them.

427 Three features of the construction motivate the identification, and each
428 distinguishes it from a bare synchrony measure.

429 First, compatibility rather than agreement. Oscillator phases can agree
430 everywhere while the local probabilistic descriptions they carry remain in-
431 compatible on their overlaps, and Section 5.2 exhibits that configuration
432 inside the formalization, not only as an illustration. If the unity of experi-
433 ence is a matter of local contents being about one world rather than merely
434 co-timed, then compatibility, not coherence, is the property to look for—and
435 coherence enters as the physical condition under which compatibility can be
436 maintained, which is what E78 asserts.

437 Second, uniqueness follows from the construction rather than from a sep-
438 arate selection principle. Accounts of conscious unity need something that
439 picks one experience from among the candidates a system could support;
440 integrated information theory postulates exclusion for this purpose (Tononi
441 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
442 whose local sections agree, the sheaf condition gives exactly one global sec-
443 tion, and given a contracting self-prediction map, Banach’s theorem gives
444 exactly one fixed point. The scope of that claim needs stating exactly, be-
445 cause the stronger reading is available and is not ours. Uniqueness is relative
446 to a cover and a map, and choosing them—which regions, which overlaps,

447 which encoding region, and so where the subject ends—is not supplied by ei-
448 ther theorem. It is what E78 and E89 carry. What the construction removes
449 is the need for an additional principle to select among the experiences a fixed
450 cover and map would otherwise admit; it does not remove the need to say
451 which cover and map a given subject has. That is a structural argument,
452 not evidence.

453 Third, the reflexive clause makes the self a condition rather than an addi-
454 tion. The predictive map restricts a global state to a region encoding it and
455 reads that encoding back out, so a fixed point is a state whose self-model is
456 accurate about the state it models. That condition is available only because
457 the state modelled is a single glued object: a system whose local descrip-
458 tions do not cohere has no such state for a self-model to be about. Unity
459 and selfhood are therefore one construction read at two levels, which is why
460 `UnifiedSelf` requires the fixed point to be the glued section itself rather
461 than an unrelated section.

462 The costs belong with the argument. The commitment does not address
463 why any physical condition is accompanied by experience; it is a proposal
464 about which physical state is the conscious one, not about why states of that
465 kind feel like anything. It is therefore open to the objection that a system
466 could satisfy every condition—compatible local states, a supercritical field,
467 an accurate self-model—while nothing is experienced. We have no reply to
468 that objection and claim none. The identification is also multiply realizable
469 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex

470 or to electromagnetic coupling, so the substrate claim is carried entirely by
471 E56 and E67 and is empirical. No conclusion about artificial systems follows
472 from the formal results.

473 What could count against the commitment is dissociation. It predicts that
474 the two clauses and the presence of a unified conscious episode go together,
475 so a state in which local cortical descriptions are demonstrably compatible
476 and the field supercritical, while the subject reports no unified experience,
477 would tell against it; so would unified reported experience where compati-
478 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
479 states are the places to look, because they separate coherence from report
480 in ways ordinary waking does not. Such tests need an operational measure
481 of overlap compatibility on neural data, which this article does not supply;
482 constructing one is the main obstacle to making the commitment empirically
483 live rather than merely stated. Supplemental Material specifies what such an
484 observable would have to satisfy—a restriction map, a statistic with a null
485 under decoding noise, validation against configurations whose compatibility
486 is built rather than inferred—and the three ways it would fail silently, one
487 of which is reading compatibility off synchrony and so assuming away the
488 distinction being tested.

489 The physical tests of Sections 3 and 4 are independent of all of this. They
490 concern coupling, coherence and recovery, and can fail while the identification
491 stands, or succeed while it is wrong.

492 7 Discussion

493 The framework connects thermodynamic information constraints, collective
494 phase organization and compatible local representations, conditionally and
495 with its assumptions named. Its biological content is a coupling-gated re-
496 covery hypothesis with defined observables, a stated protocol, and failure
497 conditions declared in advance. Its formal content is a composition whose
498 eight gaps are exhibited rather than argued past, and which can be inhabited
499 by a single model. Its interpretive content is the identification of Section 6,
500 offered as a commitment with reasons.

501 The simulations are useful because they restrict the hypothesis rather
502 than support it. Finite ramps smear the stationary onset to the point where
503 a fitted onset exponent chiefly reports the ramp; an unpinned collective
504 phase makes independence diagnostics ambiguous in exactly the supercritical
505 regime the model cares about; and squared-drift plasticity maintains coher-
506 ence while degrading structural alignment. The second and third of these
507 are results about method and about a common intuition, and they transfer
508 beyond this framework.

509 The proposal differs in emphasis from accounts based on integrated in-
510 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
511 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
512 candidate interaction medium and a mathematical compatibility condition,
513 and, relative to a given cover and self-prediction map, it derives the unique-

514 ness those accounts postulate. It does not derive the exclusion principle of
515 integrated information, identify every broadcast with a field transition, or
516 derive a generative model from a dissipation bound. Continuous fields can
517 carry incompatible local descriptions and discrete models can admit sheaf
518 constructions, so continuity alone does not settle the combination problem;
519 compatibility is what this account offers in its place.

520 Two formal results bear on the substrate question, and both admit a
521 stronger reading than they support. At matched coupling resource, a fixed
522 wiring support that excludes a better-correlated pair can be suboptimal—a
523 statement about constrained optimization, not about biology. On an atom-
524 less substrate, a kernel supported on finitely many points contributes zero to
525 the chosen continuum coupling integral. That is a property of the measure
526 and the observable, and it cuts against this framework’s own coarse-graining
527 step as much as against anything else. It does not imply that digital hard-
528 ware cannot approximate a field or support consciousness, and no biological-
529 versus-artificial conclusion follows.

530 The empirical priorities follow directly from Section 3.2: estimate effec-
531 tive coupling and diffusion in consistent units, test the proposed relation
532 between coupling and extracellular geometry, and measure recovery against
533 an independently defined behavioral endpoint under matched model compar-
534 ison. A favorable concentration–coherence relation would support a statis-
535 tical description, not identify its physical cause; an unfavorable result in a
536 regime satisfying the stated assumptions would constrain that description.

537 Successful synchronization simulations cannot substitute for measurements
538 distinguishing ephaptic interactions from other coupling mechanisms.

539 The article therefore asks to be judged in three separable parts: a pre-
540 diction that can be tested, a composition that can be checked, and an iden-
541 tification that can be argued about. Keeping them separate is what makes
542 the whole assessable.

543 **Data and code availability**

544 The Lean development, simulation scripts, tests and saved numerical sum-
545 maries are maintained in the manuscript source repository. Publication sim-
546 ulation values are regenerated from saved summaries without rerunning pro-
547 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
548 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
549 in Supplemental Material. The source repository contains the reproducible
550 analysis and does not redistribute the raw recordings.

551 **Use of AI tools**

552 AI tools assisted simulation-code development and manuscript revision. Ma-
553 chine checking establishes the stated Lean results under their hypotheses;
554 it does not validate biological interpretations or replace scientific review of
555 generated code and prose.

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